# 6 Batch reactor models[1].

## 6.0 Introduction

Chemical reactors are one of the primary unit operations in many chemical process plants. For any chemical reactor, it is necessary to be able to predict how quantities of products are formed, how fast the products are produced and the reagents are consumed. Every chemical reaction is accompanied by the release or intake of energy and we need to quantify the rate at which this occurs in order to design for safe operation. In this chapter, we develop the mathematical equations for an isothermal and non-isothermal batch reactor. The models we develop are general, in that we consider several concurrent reactions and chemical species.

## 6.1 Background

A batch reactor (see Figure 6.1) is a vessel which takes a charge of reactants which react over a suitable time period to produce products which are then discharged. The vessel can have a loose fitting cover (reaction kettle) or a cover bolted to the vessel to allow pressurised operation. The reactor is equipped with agitators to enable good mixing. As reactions may be exothermic / endothermic batch reactors are equipped with heating / cooling systems in order to maintain (control) the reaction temperature to the desired level (set-point). Batch reactors are normally used for liquid phase reactions, but may be used for liquid / solid or liquid with a gas being bubbled into the reacting mass. They are generally used for the production of high value added fine chemicals and pharmaceuticals (because of the high costs associated with their operation, loss of production when changing batches etc.). They generally require a smaller capital investment (compared to continuous reactors) especially if production is for a low tonnage (value added) chemical. Moreover, they have inherent kinetic advantages over continuous reactors for some reactions (primarily those with slow rate constants).

A typical design aim, for a batch reactor, would be to find the time at which the maximum amount of product species is produced and optimise reactor performance in terms of batch reaction time and operating temperature. To do this, the dynamic response of the batch reactor must be modelled, i.e. how the moles of the chemical species change with respect to time as well as any change in temperature as a result of the chemical reactions.

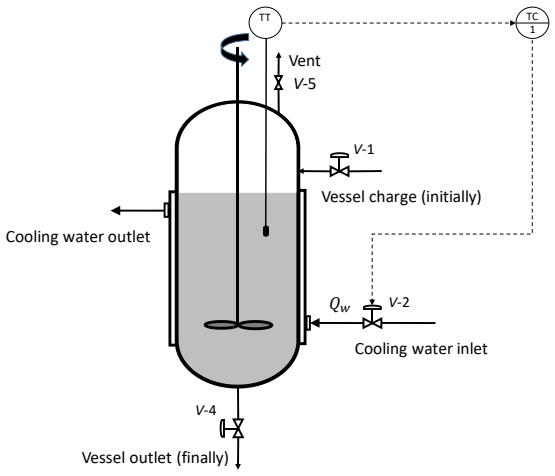



Figure 6.1. Batch reactor, with a jacketed cooling system.

To develop our mathematical models, we will assume liquid phase reactions. Furthermore, we assume that the vessel is

[1] Mark J. Willis (mark.willis@ncl.ac.uk)

perfectly mixed, i.e. the concentration of the species within the fluid are uniform (a lumped parameter description).

## 6.2 Overall mass balance

The principle of conservation of mass when applied to a dynamic system is,

**The rate of accumulation of mass**
**= the mass inflow – the mass outflow**

As there is no inflow or outflow from the reactor, the total mass balance, after the reactor has been charged, and the batch cycle has begun is therefore,

$$\frac{d(\rho V)}{dt} = 0 \qquad 6.1$$

In equation 6.1, we have, $V$ – volume of the liquid $(\mathrm{m}^3)$; $\rho$ – density of the liquid $(\mathrm{kg.\,m}^{-3})$. If the density of the liquid remains constant, then this may be written as,

$$\rho \frac{d(V)}{dt} = 0 \qquad 6.2$$

In other words, the volume of the reacting liquid is constant. More generally, we assume that the density of the liquid can vary as the reactions progress giving,

$$\rho \frac{dV}{dt} + V \frac{d\rho}{dt} = 0 \qquad 6.3$$

### 6.2.1 Volume of liquid in vessel

If we assume additive volumes, then we can calculate the volume of the reaction mass at any particular time as,

$$V = \sum_{j=1}^{N_S} \left( \frac{m_j}{\rho_j} \right) = \sum_{j=1}^{N_S} \left( \frac{n_j M_j}{\rho_j} \right) \qquad 6.4$$

In equation 6.4, we have $n_j$ - moles of the chemical species $(\mathrm{kmol})$; $M_j$ – molecular weight of species, $j$ $(\mathrm{kg\ kmol}^{-1})$; $\rho_j$ – density of species, , $j$ $(\mathrm{kg\ m}^{-3})$; $N_S$ – the total number of species.

## 6.3 Species component balances

Unlike mass, chemical components are not conserved. If a reaction occurs inside a system, the number of moles of the individual species will either increase or decrease as a result of the chemical reactions. A general component balance expression for the $j^{th}$ chemical species is,

**The rate of accumulation of moles of species, *j***
**= the molar inflow of species, *j***
**– the molar outflow of species, *j***
**+ the nett rate of production of species, *j***

For the batch reactor, as there is no inlet or outlet flow, the species component balances may be written,

$$\frac{dn_j}{dt} = VR_j \qquad 6.5$$

In equation 6.5, we have $R_j$ – the nett rate of production of chemical species $(\mathrm{kmol\ m}^{-3}\mathrm{s}^{-1})$, which can be either a positive or negative quantity.

When multiple chemical reactions occur in a system, the rates of production / consumption of the chemical species will be dependent upon the species participation in the chemical reactions as well as the individual chemical reaction rates.

## 6.4 Species participation in multiple reactions

Suppose that we have the following species, $x_1, \ldots, x_N$ and subsets of these species participate in a number of concurrent reactions, $N_R$. If we define a vector, $\mathbf{B} = (x_1, \ldots, x_N)$, which contains the names or symbolic representation of the species names, i.e., we may have $\mathbf{B} = (H_2O, \ldots, H_2)$, then for each of the chemical reactions that are occurring we have,

$$\mathbf{BS}_R \rightarrow \mathbf{BS}_P \qquad 6.6$$

In equation 6.6 we have, $\mathbf{S}_R$, the stoichiometric numbers of the reactant species in each of the reactions and, $\mathbf{S}_P$, the stoichiometric numbers of the product species in each of the reactions. Both $\mathbf{S}_R$ and $\mathbf{S}_P$ are in general, matrices, where $\mathbf{S}_{Rj,i}$ is the stoichiometric number of the $j^{th}$ (reactant) species within $\mathbf{B}$ participating in the $i^{th}$reaction and $\mathbf{S}_{Pj,i}$ is the stoichiometric number of the $j^{th}$ (product) species within $\mathbf{B}$ participating in the $i^{th}$reaction. For example, suppose we had the following chemical reactions,

$2x_1 \rightarrow x_2$

$x_1 \rightarrow x_3 \rightarrow x_4$

Using equation 6.6, we can represent the chemical reactions as,

$$(x_1 \quad x_2 \quad x_3 \quad x_4)\begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow$$

$$(x_1 \quad x_2 \quad x_3 \quad x_4)\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

## 6.5 Chemical reaction rates

Chemical reaction rates are, in general, nonlinear functions of species concentrations, $c_j$. For chemical reactions occurring in well-mixed ideal solution, a description of reaction rate is provided by the law of mass action kinetics (MaK). This states that the reaction rate of the $i^{th}$ chemical reaction, $r_i$, is proportional to the product of the concentration of the reactants raised to the power of their stoichiometric amounts,

$$r_i = k_i \prod_{j=1}^{N_S} c_j^{s_{R_{ij}}} \qquad 6.6$$

In equation 6.6, we have, $k_i$ – (isothermal) rate constant $[\mathrm{m}^3/\mathrm{kmol}]^{n-1}s^{-1}$; $c_j$ – concentration of species, $j$ $(\mathrm{kmol\ m^{-3}})$; $s_{R_{ij}}$ – reactant stoichiometric number and; $r_i$ – reaction rate $(\mathrm{kmol\ m^{-3}s^{-1}})$.

It may be noted that the units of the (isothermal) rate constant depend upon the order of the chemical reaction (sum of the reactant stoichiometric numbers, $s_{R_{ij}}$). This is because, reaction rates always have the units, $\mathrm{kmol\ m^{-3}s^{-1}}$.

(Isothermal) rate constants are known to be temperature dependent. The effect of temperature on the rate constant is usually found to be exponential (Arrhenius temperature dependency),

$$k_i = A_i \exp\left(\frac{-E_i}{RT}\right) \qquad 6.8$$

In equation 6.8, we have, $A_i$ – Frequency factor $[\mathrm{m}^3/\mathrm{kmol}]^{n-1}s^{-1}$; $E_i$ – activation energy $(\mathrm{kJ\ kmol^{-1}})$; $R$ – gas constant $(\mathrm{kJ\ kmol^{-1}K^{-1}})$ and; $T$ – temperature of the liquid $(\mathrm{K})$.

The activation energy, $E_i$, models the temperature dependence of the reaction rate, the larger the value of $E_i$ the larger the increase in $k_i$ with temperature, $T$. We observe that the (isothermal) rate constant increases nonlinearly with an increase in temperature; a rule of thumb being that a temperature increase of 10 $\mathrm{K}$ causes the (isothermal) rate constant and hence the reaction rate to double.

## 6.6 Rate of production / consumption of the chemical species

To specify the rate of production / consumption of each of the chemical species we must account for the stoichiometry of the chemical reactions. We therefore define a stoichiometric matrix,

$$\mathbf{S} = \mathbf{S}_P - \mathbf{S}_R \qquad 6.9$$

The elements of $\mathbf{S}$, the $s_{j,i}$ are the stoichiometric coefficients of the

$j^{th}$species in the $i^{th}$ reaction. By convention, $s_{j,i} < 0$ for a species that undergoes net consumption in a reaction (a reactant), $s_{j,i} > 0$ for a species that undergoes net production and $s_{j,i} = 0$ for a species that is either not involved in the $i^{th}$ reaction or has no net change in it. With these definitions, the rate of production (or consumption) of each of the chemical species as a result of each of the chemical reactions is,

$$R_j = \sum_{i=1}^{N_R} s_{j,i} r_i \qquad 6.10$$

The species component balances are therefore,

$$\frac{dn_j}{dt} = V \sum_{i=1}^{N_R} s_{j,i} r_i$$

## 6.7 Example one: isothermal batch reactor

We will consider the following reactions (known as the Van de Vusse reactions) occurring in a well-mixed, isothermally operated batch reactor,

$2x_1 \rightarrow x_2$

$x_1 \rightarrow x_3 \rightarrow x_4$

There are three chemical reactions. Assuming mass action kinetics, the reaction rates are,

$r_1 = k_1 c_1^2$

$r_2 = k_2 c_1$

$r_3 = k_3 c_3$

Noting the stoichiometry, the rates of production / consumption of each of the species are,

$R_0 = 0$

$R_1 = -2r_1 - r_2$

$R_2 = r_1$

$R_3 = r_2 - r_3$

$R_4 = r_3$

In the above terms we also include the rate of production / consumption of the solvent, which is zero, $R_0 = 0$.

Initial conditions are also required, i.e. how many moles of each of the species exist at time $t = 0$ (the assumed start time of the reaction). The initial moles of the chemical species may be calculated using,

$$n_j(0) = \frac{\varphi_j V(0) \rho_j}{M_j} \qquad 6.11$$

In equation 6.11, we have, $n_j(0)$ - initial moles of the chemical species $(\mathrm{kmol})$; $\varphi_j$ – initial volume fraction of the species and; $V(0)$ – initial volume of reacting liquid $(\mathrm{m}^3)$.

### 6.7.1 Simulation

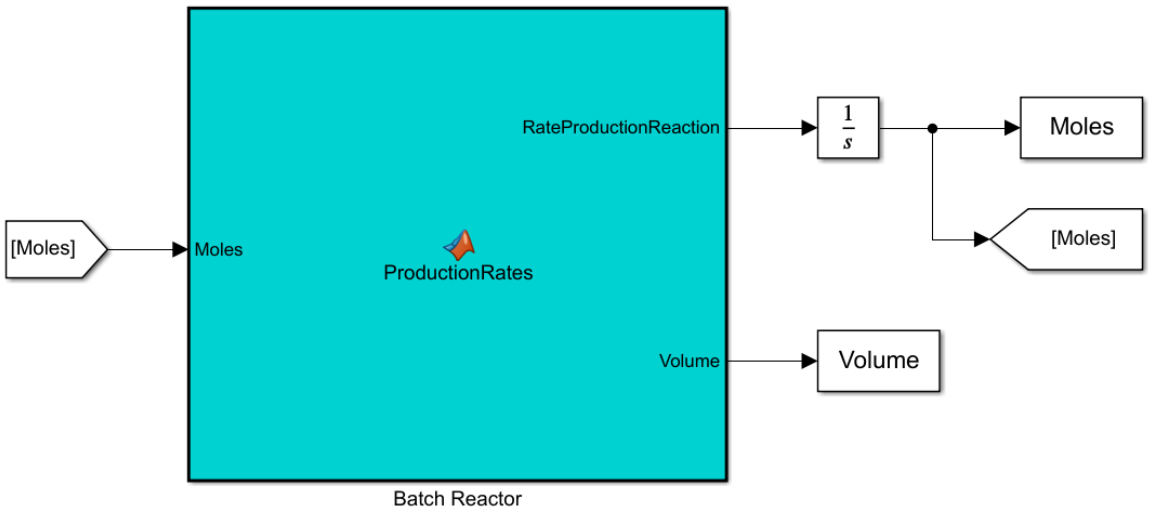
We simulate a batch reactor assuming the Van de Vusse reactions. The batch reactor is initially charged with 0.5 $\mathrm{m}^3$ of species $x_1$ (the reactant) and 0.5 $\mathrm{m}^3$ of water (the solvent, $x_0$). At, $t = 0$, the reacting mixture is at 60 C. The reactor operates for a period of two hours. Species, $x_2$ is a by-product of this reaction scheme, $x_3$, the primary product and $x_4$ a waste product.

The required model constants are given in table 6.1 and the reaction rate constants are provided in table 6.2.

| | | | |
|---|---|---|---|
| $A_1$ | 8088 $\mathrm{m^3 kmol\ s^{-1}}$ | $E_1$ | 50434 $\mathrm{kJ\ kmol^{-1}}$ |
| $A_2$ | 4.7 x $10^8$ $\mathrm{s^{-1}}$ | $E_2$ | 74650 $\mathrm{kJ\ kmol^{-1}}$ |
| $A_3$ | 8.73 x $10^8$ $\mathrm{s^{-1}}$ | $E_3$ | 60899 $\mathrm{kJ\ kmol^{-1}}$ |
| $M_0$ | 18 $\mathrm{kg\ kmol^{-1}}$ | $\rho_0$ | 1000 $\mathrm{kg\ m^{-3}}$ |
| $M_1$ | 100 $\mathrm{kg\ kmol^{-1}}$ | $\rho_1$ | 800 $\mathrm{kg\ m^{-3}}$ |
| $M_2$ | 200 $\mathrm{kg\ kmol^{-1}}$ | $\rho_2$ | 900 $\mathrm{kg\ m^{-3}}$ |
| $M_3$ | 100 $\mathrm{kg\ kmol^{-1}}$ | $\rho_3$ | 1000 $\mathrm{kg\ m^{-3}}$ |
| $M_4$ | 100 $\mathrm{kg\ kmol^{-1}}$ | $\rho_4$ | 1000 $\mathrm{kg\ m^{-3}}$ |

Table 6.1. Parameters for the isothermal batch reactor

A script file initialises and then simulates the isothermal batch reactor model. The Simulink implementation of the isothermal batch model uses an embedded MATLAB function to code the required rate terms.




As input to the embedded function, we have the process data initialised in the MATLAB script file as well as the current value of the moles of each of the chemical species. The embedded MATLAB function calculates the rate of change of moles of the chemical species as well as the volume of the reacting fluid (assumed to be a liquid). We use an integrator block to integrate the differential equations. The moles of the chemical species as well as the volume are passed back to the MATLAB workspace for plotting and any subsequent analysis. The simulation results are shown in Figure 6.3.

## 6.8 Stoichiometric matrix representation

A stoichiometric matrix is formed by using the stoichiometric coefficients of the species in each of the reactions that we consider. The matrix is organised so that every column corresponds to a reaction and every row a chemical species. The entries in the matrix are the stoichiometric coefficients of the species (a negative value corresponding to a reactant species in the particular reaction and a positive value a product). For the Van de Vusse reaction scheme the stoichiometric matrix is,

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 \\ -2 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \qquad 6.12$$

This allows the rate of change of the moles of the chemical species to be expressed in terms of stoichiometry and $N_R$ individual reaction rates in matrix-vector notation,

$$\frac{\mathrm{d}\mathbf{n}}{\mathrm{dt}} = V\mathbf{S}\mathbf{r}, \quad \mathbf{n}(t_0) = \mathbf{n}_0 \qquad 6.13$$

The ODEs describe the rate of change of the moles, $\mathbf{n}$ $(N_S \mathrm{x}\ 1)$, of the chemical species with respect to time, t. $\mathbf{r}$ $(N_R \mathrm{x}\ 1)$ comprises the $N_R$ reaction rate terms, $V$ the volume, $\mathbf{n}_0(N_S \mathrm{x}\ 1)$ the initial conditions at time, $t_0$.

## 6.9 Example one: analytical solutions

For chemical reactions, where the chemical reaction rates are a linear function of the species concentrations, we can obtain analytical solutions to the isothermal batch reactor design equations. We demonstrate this for the reversible reaction scheme,

$x_1 \rightleftharpoons x_2$

The chemical reaction rates are assumed to be, $r_1$ for the reaction $x_1 \longrightarrow x_2$ and $r_2$

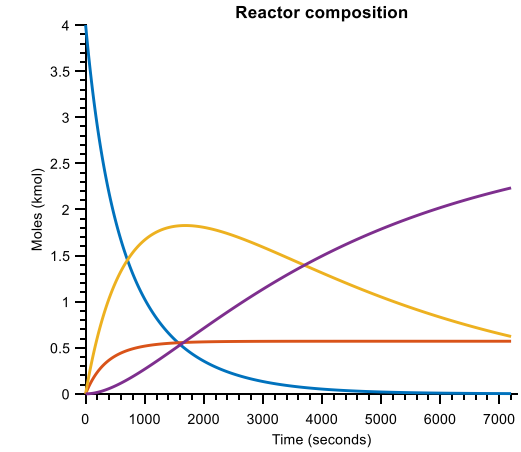
for the reaction $x_2 \longrightarrow x_1$; the rates are given by,

$$r_1 = k_1 c_1$$

$$r_1 = k_2 c_2$$

For the given reaction scheme, the stoichiometric matrix is,

$$\mathbf{S} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$



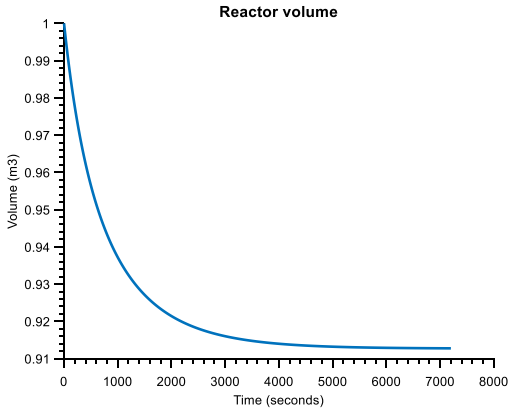




**Figure 6.3.** Species moles and volume as a function of time (Van de Vusse kinetics). We observe that the reactant species exponentially decays to zero. The species of interest is the intermediate that passes through a maximum at around 1500 seconds; this would be the ideal time to end the batch as the maximum amount of product is produced. Note that, as the reactions proceed volume decreases as a result of the change in density.

Using equation 6.13 and assuming constant volume, gives, two differential equations describing the rate of each of the concentration of the two species,

$$\begin{pmatrix} \frac{dc_1}{dt} \\ \frac{dc_2}{dt} \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} k_1 c_1 \\ k_2 c_2 \end{pmatrix} = \begin{pmatrix} -k_1 & k_2 \\ k_1 & -k_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad 6.14$$

To obtain the analytical solution we can use Laplace transforms. Taking Laplace transforms of equation 6.14, we have,

$$s \begin{pmatrix} c_1(s) \\ c_2(s) \end{pmatrix} - \begin{pmatrix} c_1(0) \\ c_2(0) \end{pmatrix} = \begin{pmatrix} -k_1 & k_2 \\ k_1 & -k_2 \end{pmatrix} \begin{pmatrix} c_1(s) \\ c_2(s) \end{pmatrix} \quad 6.15$$

This can be re-arranged to give,

$$\begin{pmatrix} c_1(s) \\ c_2(s) \end{pmatrix} = \begin{pmatrix} s + k_1 & -k_2 \\ -k_1 & s + k_2 \end{pmatrix}^{-1} \begin{pmatrix} c_1(0) \\ c_2(0) \end{pmatrix} \quad 6.16$$

Calculating the matrix inverse and assuming $c_2(0) = 0$, gives the following two equations,

$$c_1(s) = \frac{s + k_2}{s^2 + (k_1 + k_2)s} c_1(0)$$

$$c_2(s) = \frac{k_1}{s^2 + (k_1 + k_2)s} c_1(0)$$

If we take inverse Laplace transforms (partial fraction expansion is required) we obtain the analytical solutions,

$$c_1(t) = e^{-(k_1+k_2)t} c_1(0) + \frac{k_2}{k_1+k_2}\left(1 - e^{-(k_1+k_2)t}\right) c_1(0)$$

$$c_2(t) = \frac{k_1}{k_1 + k_2}\left(1 - e^{-(k_1+k_2)t}\right) c_1(0)$$

## 6.10 Energy balance

Assuming constant pressure, the enthalpy balance is,

$$\frac{d(m\hat{H})}{dt} = q \quad 6.17$$

The first term in equation 6.17 represents the change in enthalpy of the reacting

system with respect to time while $q$ is the energy that flows in (or out) of the system due to heat transfer at the interface.

As we have reactions, our preferential basis for calculation is moles, therefore we consider the specific molar enthalpy and, neglecting any enthalpy of mixing, re-write the enthalpy balance as,

$$\frac{d\left(\sum_{j=1}^{N_s} n_j \bar{H}_j\right)}{dt} = q \qquad 6.18$$

In equation 6.18, we have, $\bar{H}_j$ – specific molar enthalpy $(\mathrm{kJkmol^{-1}})$. Notice, that there is no term saying 'rate of heat generation by reaction' as this is indirectly included within the specific molar enthalpy.

Differentiating the left hand side of equation 6.18 using the chain rule gives,

$$\left(\sum_{j=1}^{N_s} n_j \frac{d\bar{H}_j}{dt}\right) + \left(\sum_{j=1}^{N_s} \bar{H}_j \frac{dn_j}{dt}\right) = q \qquad 6.19$$

The specific molar heat capacity is defined as,

$$\frac{\partial \hat{H}_j(T)}{\partial T} = \hat{c}_{p,j}(T) \qquad 6.20$$

In equation 6.20, we have, $\hat{c}_{p,j}$ – specific molar heat capacity $(\mathrm{kJkmol^{-1}K^{-1}})$. This allows the energy balance equation to be written as,

$$\left(\sum_{j=1}^{N_s} n_j\, \hat{c}_{p,j}(T)\right)\frac{dT}{dt} + \left(\sum_{j=1}^{N_s} \bar{H}_j \frac{dn_j}{dt}\right) = q \qquad 6.21$$

The component balance expression for chemical reactions in a batch reactor is,

$$\frac{dn_j}{dt} = VR_j = V\sum_{i=1}^{N_R} s_{j,i} r_i \qquad 6.22$$

Substituting this into the energy balance equation gives,

$$\left(\sum_{j=1}^{N_s} n_j\, \hat{c}_{p,j}(T)\right)\frac{dT}{dt} + V\left(\sum_{j=1}^{N_s} \bar{H}_j \sum_{i=1}^{N_R} s_{j,i} r_i\right) = q \qquad 6.23$$

By definition, the enthalpy change of the $i^{th}$ chemical reaction is,

$$\Delta H_{Ri} = \sum_{j=1}^{N_s} s_{j,i}\, \bar{H}_j \qquad 6.24$$

We can re-order the summation terms in equation 6.23 to give,

$$\left(\sum_{j=1}^{N_s} \bar{H}_j \sum_{i=1}^{N_R} s_{j,i} r_i\right) = \sum_{j=1}^{N_s}\sum_{i=1}^{N_R} s_{j,i}\bar{H}_j r_i = \sum_{i=1}^{N_R}\sum_{j=1}^{N_s} s_{j,i}\bar{H}_j r_i \qquad 6.25$$

Using equation 6.25 the final form of the energy balance is,

$$\left(\sum_{j=1}^{N_s} n_j\, \hat{c}_{p,j}(T)\right)\frac{dT}{dt} = -V\sum_{i=1}^{N_R} \Delta H_{Ri} r_i + q \qquad 6.26$$

The term, $q_R = V\sum_{i=1}^{N_R} \Delta H_{R,i} r_i$ quantifies the rate of heat evolution by the chemical reactions. In order to maintain an isothermal condition, the rate of heat evolution must be compensated for by the cooling system. Recall, that $q$ represents the energy that flows in (or out) of the system due to heat transfer at the interface. If the vessel were assumed to

have a cooling jacket then an appropriate heat transfer relationship gives,

$$\left(\sum_{j=1}^{N_s} n_j\, \hat{c}_{p,j}(T)\right)\frac{dT}{dt} \qquad 6.27$$
$$= -V\sum_{i=1}^{N_R}\Delta H_{Ri} r_i + UA_J(T_J - T)$$

### 6.10.1 Enthalpy of reaction

If the enthalpy of reaction cannot be assumed to be constant, for example, if there are significant changes in the reaction temperature, or the enthalpy of reaction data is only reported at standard conditions (normally 298.15 K and 1 atm.) then we can account for this variation. We note, from the specific molar enthalpy that, $\bar{H}_j$, is the standard heat of formation, $\left(\Delta h_j^f\right)_{Tref}$ of species, $j$, from its elements in the standard state.

To obtain the enthalpy at a temperature, $T$, we have,

$$\bar{H}_j = \left(\Delta h_j^f\right)_{Tref} + \int_{Tref}^{T} \hat{c}_{p,j}(T)dT \qquad 6.28$$

Therefore,

$$\sum_{i=1}^{N_R}\sum_{j=1}^{N_s} s_{ij}\bar{H}_j r_i = \sum_{i=1}^{N_R}\sum_{j=1}^{N_s} s_{ij}\left(\left(\Delta h_j^f\right)_{Tref} + \int_{Tref}^{T}\hat{c}_{p,j}(T)dT\right)r_i$$

Which gives,

$$\sum_{i=1}^{N_R}\sum_{j=1}^{N_s} s_{j,i}\bar{H}_j r_i = \sum_{i=1}^{N_R}(\Delta H_{Ri})_{Tref} r_i + \sum_{i=1}^{N_R}\sum_{j=1}^{N_s} s_{j,i}\left(\int_{Tref}^{T}\hat{c}_{p,j}(T)dT\right)r_i$$

### 6.11 Limiting cases

**6.11.1 Adiabatic operation** – there is no heat transfer to the surroundings (i.e. $U = 0$). Therefore,

$$\left(\sum_{j=1}^{N_s} n_j\, \hat{c}_{p,j}(T)\right)\frac{dT}{dt} = -V\sum_{i=1}^{N_R}\Delta H_{Ri} r_i$$

**6.11.2 Isothermal operation** – the temperature does not change with respect to time. Therefore,

$$V\sum_{i=1}^{N_R}\Delta H_{Ri} r_i = UA_J(T_J - T)$$

This algebraic may be used to determine the required jacket temperature to ensure isothermal operation.

### 6.12 Example two: a non-isothermal batch reactor

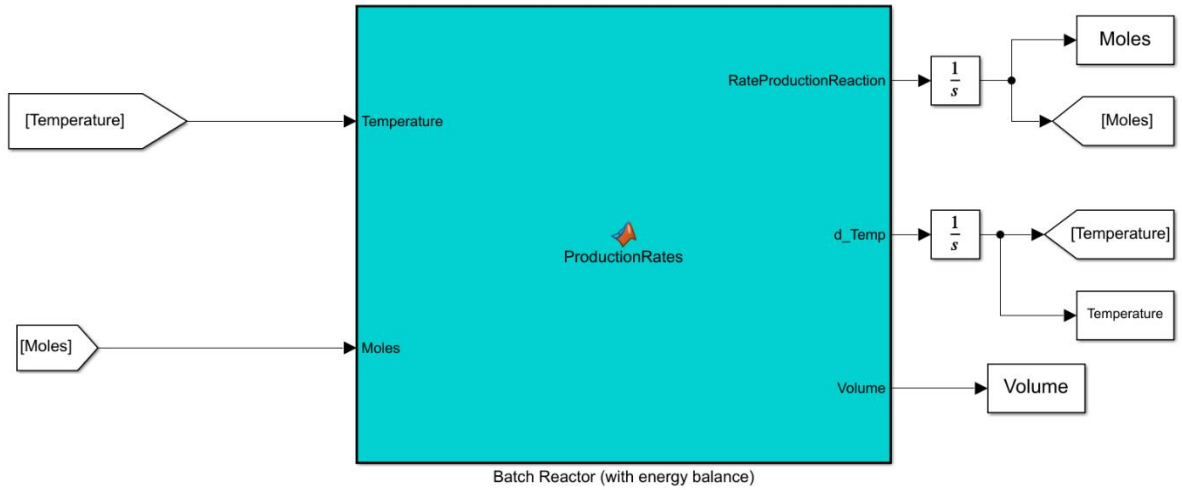
We consider the same example as studied previously – the Van de Vusse reactions and add to this the energy balance equation 6.27. We assume a constant jacket temperature, $T_J$. The data used in the simulation is summarised in Tables 6.1, 6.2 and 6.3.

| | Rate | A | E | $\Delta H_R$ |
|---|---|---|---|---|
| $2x_1 \rightarrow x_2$ | $k_1 c_1^2$ | 8088 | 50434 | -8e4 |
| $x_1 \rightarrow x_3$ | $k_2 c_1$ | 4.7e8 | 74650 | -1e4 |
| $x_3 \rightarrow x_4$ | $k_3 c_3$ | 8.73e8 | 60899 | -1e5 |

**Table 6.2.** Chemical reactions, reaction rates. The rate expressions are developed using the assumption of mass action kinetics. A and E are the Arrhenius temperature dependency terms, $\Delta H_R$ the heats of reaction.

| Name | Identifier | $M_j$ | $\rho_j$ | $c_{pj}$ |
|---|---|---|---|---|
| Solvent | $x_0$ | 18 | 1000 | 4.2 |
| Reagent | $x_1$ | 100 | 800 | 2 |
| By-product | $x_2$ | 200 | 900 | 2 |
| Product | $x_3$ | 100 | 1000 | 2 |
| Waste Product | $x_4$ | 100 | 1000 | 2 |

**Table 6.3.** Component properties (Note that the density of the components are different therefore as the reaction progresses there will be a change in volume.




We initialise the simulation using a script file to define the model parameters and initial conditions. We determine the volume, the rate of change of the moles of the species as well as the rate of change of the temperature using an embedded MATLAB function. As input to the embedded function we have the process data that is initialised in the MATLAB workspace as well as the current value of the moles of each of the chemical species and temperature. We use an integrator block to integrate the differential equations. The moles of the chemical species, temperature and the volume are passed back to the MATLAB workspace for plotting and any subsequent analysis. The simulation produces the results shown in Figure 6.3.

## 6.13 Design for stability

Of paramount importance is the design of a reactor that ensures stable as well as reliable operation. For an exothermic reaction, if the amount of heat generated by the chemical reactions cannot be removed by the cooling system then system stability cannot be assured and there is the potential for thermal run-away to occur. Looking at Figure 6.3, the results of the non-isothermal reactor simulation with a fixed jacket temperature of $T_J = 25$ C, we observe the small rise in temperature at the start of the batch run before the temperature starts to drop as the reaction proceeds. This is an indication of potential stability problems.

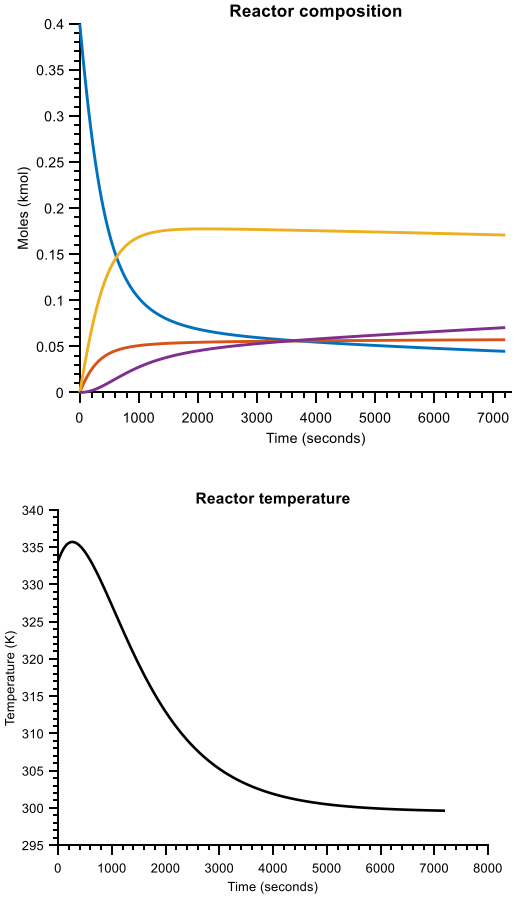



**Figure 6.3.** Species moles and temperature as a function of time (Van de Vusse kinetics). Note how the change in temperature affects the moles of the chemical species (compare with figure 3). As we are using a constant jacket temperature, we have a slight rise in the temperature at the start of the batch followed by a slow decrease in temperature as the batch time increases.

Luyben (2007) introduced a reactor stability index, equation 6.29, which may be used to define the controllability of a reactor,

$$RSI = \frac{\text{design } \Delta T}{\text{maximum } \Delta T} = \frac{T - T_{J,R}}{T - T_{J,in}} \qquad 6.29$$

In equation 6.29, the maximum (or total available) $\Delta T$ is the difference between the reactor temperature, $T$, and the supply (or inlet) temperature of the jacket coolant, $T_{J,in}$, while the design $\Delta T$ is the difference between the reactor temperature and the jacket temperature that would be required in order to maintain isothermal operation. If the required jacket temperature, $T_{J,R}$, is greater than the jacket inlet temperature, $T_{J,in}$, then it will not be possible to control the process using a cooling jacket (or by using the chosen fluid in the jacket). The closer the $RSI$ is to one the less effective the control will be as there is very little $\Delta T$ driving force to be able to increase heat removal rates in order to control against system variations. The design of the optimal operating policy for a reactor should therefore aim to achieve a small $RSI$ to ensure could stable operation and control. For a batch reactor, we should design so that a given $RSI$ limit is not exceeded for the entire processing period.

To find the required jacket temperature, $T_{J,R}$, as well as the $RSI$ we can solve the material and energy balance equations for the batch reactor. We must add the additional constraint that,

$$\frac{dT}{dt} = 0$$

We then solve the steady-state energy balance to find, $T_{J,R}$, given a choice of $T_{J,in}$. We therefore modify the Simulink simulation introduced in section 6.2. We use an algebraic constraint block to ensure the rate of change of temperature is zero.

The algebraic constraint block calculates the required jacket temperature, $T_{J,R}$ which is an input to the embedded MATLAB function. The temperature of the reactor is specified as the initial temperature and remains at this temperature because of the adjustment to, $T_{J,R}$.

It may be observed from figure 6.4 that at the start of the batch the required temperature is lower than the jacket inlet temperature, demonstrating that effective control will not be achieved. As the batch reactor time increases (and the reactions near completion, implying significant reduction of the rate of heat evolution), the $RSI$ reduces significantly.

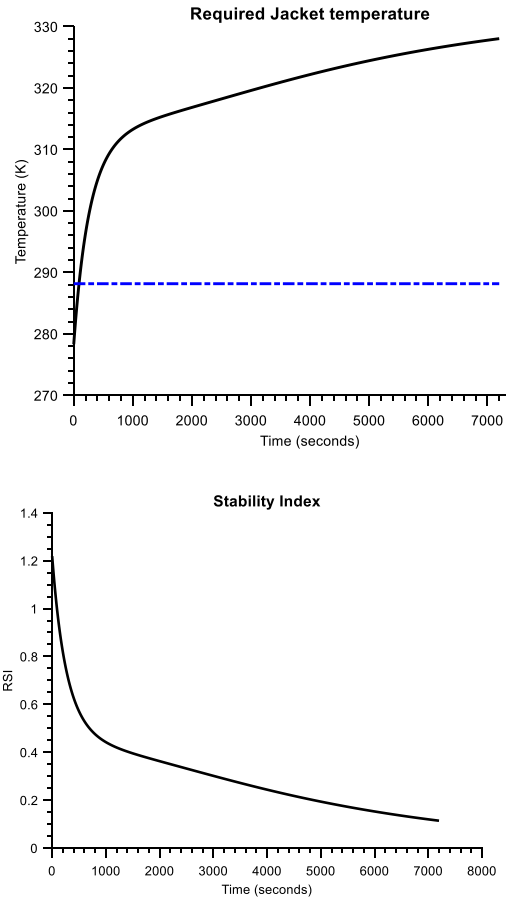
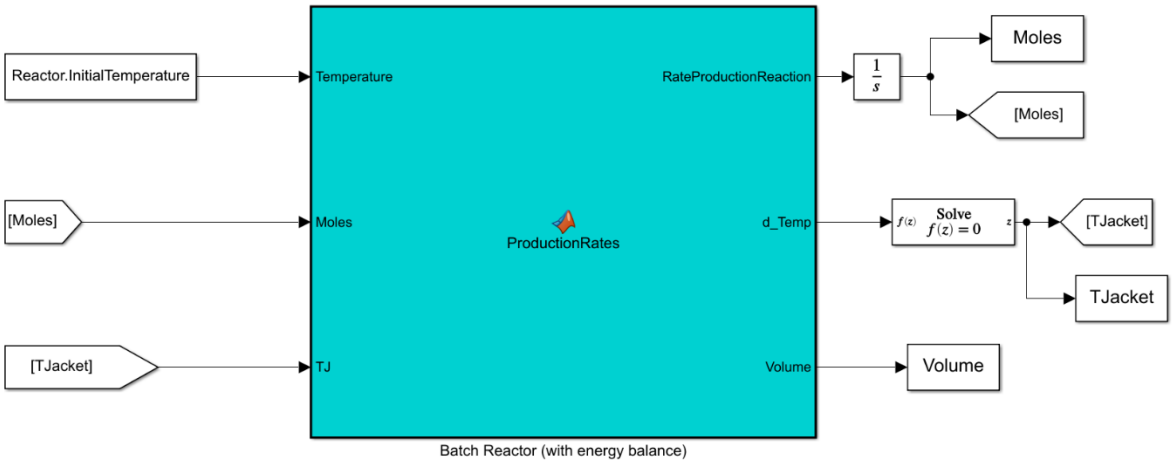



Figure 6.4. Required jacket temperature and $RSI$. A jacket inlet cooling temperature, $T_{J,in} = 15$ C is specified.




Unless the coolant is supplied at a temperature much lower than 278 K we cannot control this system at the start of the batch. Options would be, change the coolant, use an alternative means to remove heat from the reactor, e.g. cooling coils or perhaps an external heat exchanger.

## Further reading

Luyben, W.L. (2007) Chemical reactor design and control, John Wiley and Sons.

## Exercises

**Example 1**: To the isothermal batch reactor model, add calculations (code) to assess the conversion of the reactant species as well as the selectivity and yield of the product species.

**Example 2:** To the isothermal batch reactor model, add calculations (code) to stop the simulation based upon a specified conversion of the reactant species.

**Example 3:** Using the non-isothermal model, control the temperature of the reactor. Configure the feedback controller to adjust the jacket temperature in order to maintain the reactor vessel temperature. Add a step increase in the reactor temperature at the midpoint of the simulation. Note; in reality, we would adjust the rate at which a coolant enters the jacket of the reactor. We will consider the modelling of jacketed vessels in the next chapter.

## Appendix 1 Mass balance relationships for batch chemical reactors

For a batch reactor we have,

$$\frac{d(\rho V)}{dt} = 0$$

Therefore,

$$\rho \frac{dV}{dt} + V \frac{d\rho}{dt} = 0$$

Assuming additive volumes, and the density of each species, $\rho_j$, is constant with respect to temperature (and pressure) we have,

$$V = \sum_{j=1}^{N_s} \frac{n_j M_j}{\rho_j}$$

Similarly, density may be defined as,

$$\rho = \frac{m}{V} = \frac{\sum_{j=1}^{N_s} n_j M_j}{\sum_{j=1}^{N_s} \frac{n_j M_j}{\rho_j}} = \frac{\sum_{j=1}^{N_s} n_j M_j}{V}$$

Working out the derivatives we have,

$$\frac{dV}{dt} = \sum_{j=1}^{N_s} \frac{dn_j}{dt} \frac{M_j}{\rho_j}$$

$$\frac{d\rho}{dt} = \frac{\left(\sum_{j=1}^{N_s} \frac{dn_j}{dt} M_j\right) V - \left(\sum_{j=1}^{N_s} n_j M_j\right) \frac{dV}{dt}}{V^2}$$

Substituting, we obtain,

$$\frac{\sum_{j=1}^{N_s} n_j M_j}{V} \frac{dV}{dt} + V \left( \frac{\left(\sum_{j=1}^{N_s} \frac{dn_j}{dt} M_j\right) V - \left(\sum_{j=1}^{N_s} n_j M_j\right) \frac{dV}{dt}}{V^2} \right) = 0$$

Simplifying gives,

$$\left( \sum_{j=1}^{N_s} \frac{dn_j}{dt} M_j \right) = 0$$

Substituting for the rate of change of moles for a batch reactor gives,

$$V \sum_{j=1}^{N_s} R_j M_j = 0$$

The summation is, by definition, zero (it is a conservation relationship).